

**EMOTION-AWARE AGILE SYSTEM FOR PROGRESS
TRACKING, EXPERTISE RECOMMENDATION,
REQUIREMENT MANAGEMENT AND INCLUSIVE
COMMUNICATION**

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Project Proposal Report

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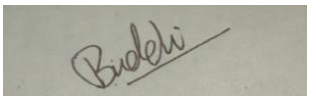
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Declaration

We declare that this is our work, and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or Institute of higher learning, and to the best of our knowledge and belief, it does not contain any material previously published or written by another person except where the acknowledgment is made in the text.

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The supervisor/s should certify the proposal report with the following declaration. The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the Supervisor: (Ms. Ishara Weerathunga)	Date
	30/8/2025.

Abstract

Contents

Declaration

Abstract

1.0 Background & Literature Survey

1.1 Research Gap

1.2 Research Problem

2.0 Objectives

Main Objectives

Specific Objectives

2.2.1 Implicit Material Requirement Prediction

2.2.2 Dynamic Supply Chain Optimization

2.2.3 Predictive Waste Optimization and Cross-Project Learning

3.0 Methodolog

4.0 Project Requirements

4.1 Functional Requirements

4.2 Non-Functional Requirements

4.3 System Requirements

Introduction

Effective communication is one of the most critical elements in Agile software development, as team success depends heavily on collaboration, knowledge sharing, and active participation during meetings. However, many teams face recurring challenges where valuable ideas are lost due to communication barriers such as shyness, poor confidence, fragmented English, or hesitant speech. Research has shown that existing brainstorming tools often favor outspoken participants and do little to empower less confident contributors, leading to bias in idea generation and decision-making [1][2].

Recent studies highlight the potential of artificial intelligence (AI) in supporting collaborative brainstorming, where AI agents can refine raw contributions, provide alternative perspectives, and structure group discussions [1][3]. However, existing platforms such as Miro, Jamboard, and even AI-powered assistants like Grammarly [7] or DeepL Write [8] primarily focus on capturing ideas or correcting grammar, without addressing inclusivity, hesitation, or confidence-building. This creates a gap where many Agile team members—especially those with limited English proficiency or communication anxiety—remain underrepresented.

This project proposes the development of an **Inclusive Brainstorming and Communication Empowerment Platform**, a standalone AI-powered system that transforms incomplete, grammatically incorrect, or hesitant contributions into clear, professional Agile-style suggestions. The system incorporates features such as silent idea submission, hesitation detection, entity-preserving NLP rephrasing, tone adjustment, and confidence-building previews, ensuring every team member can contribute meaningfully to brainstorming sessions. Unlike conventional tools,

it also introduces **AI clustering and summarization of ideas** to reduce redundancy, and **participation diversity analytics** to quantify inclusivity during Agile meetings.

By combining insights from NLP, HCI, and software engineering, this research seeks to address inclusivity and fairness challenges in Agile brainstorming. Prior work in multilingual NLP [4][5][6] demonstrates the feasibility of handling low-resource or mixed-language inputs, further supporting the platform’s potential extension for diverse teams. Ultimately, the system aims to unlock hidden contributions, improve clarity, and ensure structured, fair, and effective collaboration in Agile software development teams.

1.0 Background & Literature Review

The role of effective communication in Agile software development has gained significant attention in recent years. Agile methodologies emphasize collaboration, adaptability, and frequent feedback, but their success depends largely on the ability of all team members to contribute effectively during brainstorming sessions, sprint planning, and retrospectives. However, existing studies highlight recurring challenges where valuable contributions are lost due to communication barriers such as shyness, fragmented English, low confidence, or hesitation to speak in public [1][2]. These communication challenges often lead to participation imbalances, where a small group of outspoken members dominate discussions while less confident individuals remain silent.

Traditional brainstorming and collaboration tools, such as Miro, Jamboard, or Slack, provide digital platforms to capture and visualize ideas, but they primarily serve as raw input collectors and lack mechanisms to refine, structure, or encourage contributions from hesitant participants. Existing research on human–AI collaboration in brainstorming has demonstrated that AI can act as a facilitator by refining ideas, suggesting alternatives, and guiding discussions [1][3]. Yet, these approaches largely overlook the inclusivity dimension—how to ensure that every participant, regardless of language proficiency or confidence, is empowered to contribute.

AI-driven communication assistants such as Grammarly [7] and DeepL Write [8] have proven effective in correcting grammar and improving text fluency. However, their design is primarily aimed at general-purpose writing rather than the real-time, collaborative, and Agile-specific environment of team meetings. These tools do not preserve technical entities such as

task IDs, numerical values, or domain-specific jargon, which are critical in software engineering discussions. More importantly, they do not provide confidence-boosting features such as previews or hesitation detection, leaving significant gaps in supporting shy or hesitant contributors.

Recent works on AI-assisted brainstorming emphasize structured idea generation and clustering. For instance, Köhl and Lins [3] explored how AI augmentation can improve collaborative idea generation by reducing redundancy and highlighting common themes. Similarly, empirical studies on generative AI in brainstorming [2] show that AI can support users in articulating ideas more clearly and systematically. However, most implementations stop at clustering or visualization, without addressing confidence-building or inclusivity metrics that measure fairness of participation.

There is also growing interest in multilingual and low-resource NLP research, particularly in the Sri Lankan context. Studies on Sinhala-English code-mixed text [4][5] and multilingual entity recognition [6] show promising approaches for processing fragmented or non-standard inputs. While multilingual input handling is beyond the primary scope of this project, these advances demonstrate the feasibility of integrating such features in future extensions of inclusive brainstorming systems.

Despite these advancements, current platforms remain limited in addressing the communication inequities within Agile meetings. Most systems assume fluency and confidence among contributors, overlooking the fact that many ideas are lost due to incomplete, hesitant, or grammatically flawed expression. As a result, there is a clear research gap in developing AI-powered brainstorming systems that not only refine and clarify raw contributions but also empower silent or hesitant members through confidence-building workflows, inclusivity analytics, and structured decision support.

This project builds on the existing literature by proposing an Inclusive Brainstorming and Communication Empowerment Platform, which combines NLP-driven rephrasing, hesitation detection, confidence-boosting previews, and AI clustering to create a fairer and more effective brainstorming process. By focusing on inclusivity and structured participation, this research aims to unlock hidden contributions and ensure balanced, professional communication in Agile team environments.

1.1 Research Gap

Although significant progress has been made in the integration of AI into brainstorming and communication support, existing systems reveal several key limitations. Current collaboration tools such as Miro and Jamboard provide digital spaces to capture raw ideas, but they lack the ability to refine, structure, or encourage contributions from team members who struggle with confidence or language [1][2]. These tools assume that all participants are equally fluent and expressive, which leads to biased discussions dominated by a few outspoken individuals while valuable but less confident contributions are overlooked.

Similarly, existing AI-powered writing assistants like Grammarly [7] and DeepL Write [8] focus primarily on grammar correction and language fluency in general-purpose contexts. While effective in improving textual quality, they are not designed for Agile environments and fail to preserve technical entities such as IDs, numbers, or domain-specific terminology that are critical during software development meetings. Moreover, these tools do not incorporate confidence-building features such as hesitation detection, before-and-after previews, or tone adjustment that could empower shy or hesitant contributors to participate more actively.

Recent studies on AI-assisted brainstorming [1][3] demonstrate the potential of clustering and summarizing contributions to improve decision-making efficiency. However, these approaches primarily address idea organization and redundancy, not the inclusivity and fairness of participation. No existing system measures or ensures diversity in contributions through quantifiable inclusivity metrics such as participation balance or equality indices. This leaves a critical gap in fostering equitable collaboration in Agile team settings.

Furthermore, while research in multilingual NLP [4][5][6] highlights the potential for handling code-mixed or fragmented text, few systems extend these findings to real-time team communication environments. This gap further marginalizes participants who may prefer to express themselves in non-standard English or local languages, particularly in multicultural teams.

In summary, the current literature lacks a solution that combines real-time grammar-aware rephrasing, confidence-building features, structured brainstorming workflows, intelligent clustering, and inclusivity measurement into a single platform. Addressing these limitations forms the core research gap that this project seeks to fill through the development of an Inclusive Brainstorming and Communication Empowerment Platform.

1.2 Research Problem

Agile software development relies heavily on effective communication, yet many team members face barriers such as shyness, low confidence, fragmented English, or hesitation to speak during brainstorming sessions. As a result, valuable contributions often go unheard, creating participation imbalances and biased decision-making dominated by outspoken individuals. Existing collaboration tools like Miro or Jamboard capture raw ideas but lack mechanisms to refine or encourage hesitant inputs, while AI writing assistants such as Grammarly and DeepL Write improve fluency but are not designed for Agile contexts, often failing to preserve technical details or empower contributors. This creates a significant gap in ensuring inclusivity, clarity, and fairness in Agile meetings, where silent or less confident voices remain underrepresented. The research problem, therefore, is the absence of an AI-powered system that can transform incomplete or hesitant inputs into clear, professional Agile-style contributions while providing confidence support and measurable inclusivity in brainstorming sessions.

Objectives

The “Inclusive Brainstorming & Communication Empowerment” component is to develop an AI-powered brainstorming platform that enhances Agile meetings by enabling shy, hesitant, or less confident team members to contribute effectively. The system will transform incomplete or grammatically incorrect inputs into clear, professional Agile-style suggestions, support silent or

anonymous submissions, detect hesitation, and provide confidence-building previews. In addition, it will structure brainstorming outputs through AI clustering, summarization, and inclusivity analytics to ensure balanced and fair participation across all team members.

Specific Objectives

- **Develop an AI Rephraser and Confidence-Booster Module**

This objective focuses on building a Natural Language Processing (NLP) engine that can transform raw, fragmented, or grammatically incorrect inputs into clear and professional Agile-style suggestions. The module will ensure that important technical details such as identifiers, numerical values, and domain-specific terminology are preserved during rephrasing. It will also provide a preview function to show users how their input will appear after refinement and allow them to adjust tone (e.g., polite, neutral, assertive). This ensures that contributors not only sound professional but also gain confidence in expressing their ideas.

- **Implement Silent Submission and Hesitation Detection Features**

This objective aims to create mechanisms that allow participants to submit ideas silently or anonymously through a text channel, reducing the pressure to speak publicly. Additionally, the system will include hesitation detection features, which identify signs such as prolonged typing without submission, incomplete messages, or pauses in speech input. When hesitation is detected, the system will gently prompt users with supportive suggestions, helping them express their thoughts more effectively.

- **Design a Collaborative Idea Pool with AI Clustering and Summarization**

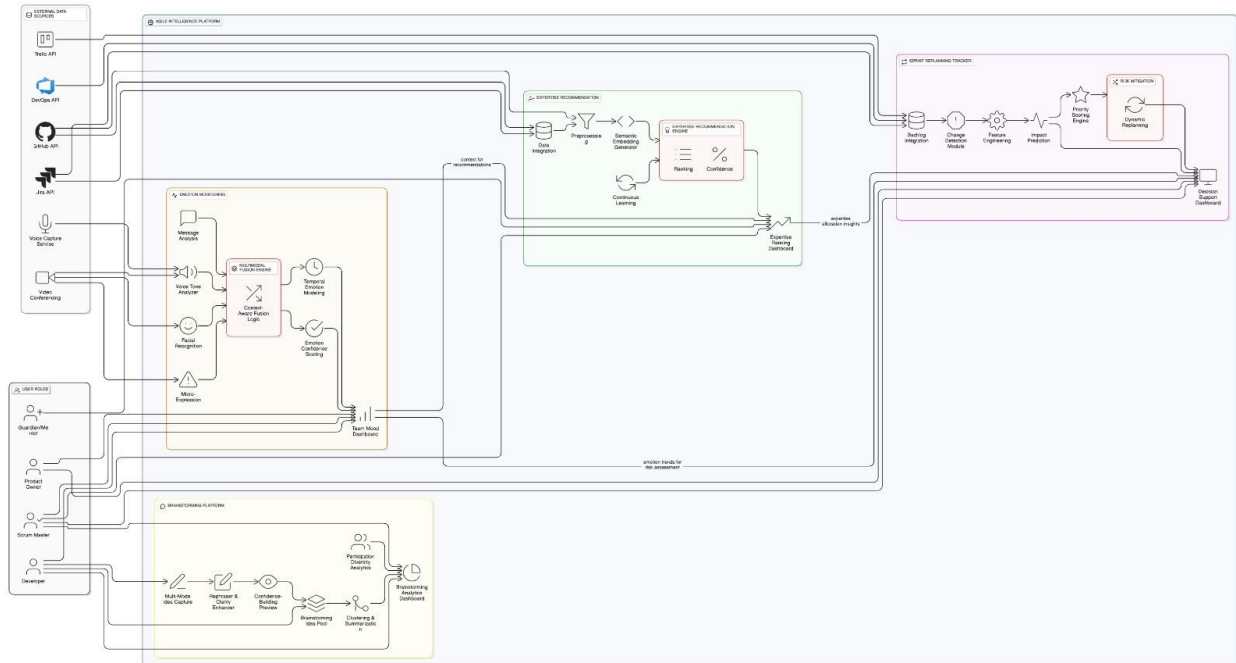
This objective focuses on structuring the brainstorming process by collecting all ideas into a shared digital workspace. Using clustering techniques, the system will group similar contributions together, eliminate redundancy, and provide automatic summaries for easier interpretation. This feature ensures that discussions are more efficient, with teams focusing on refined sets of ideas rather than repeating the same points, ultimately streamlining decision-making.

- **Develop Inclusivity Analytics and Visualization Dashboard**

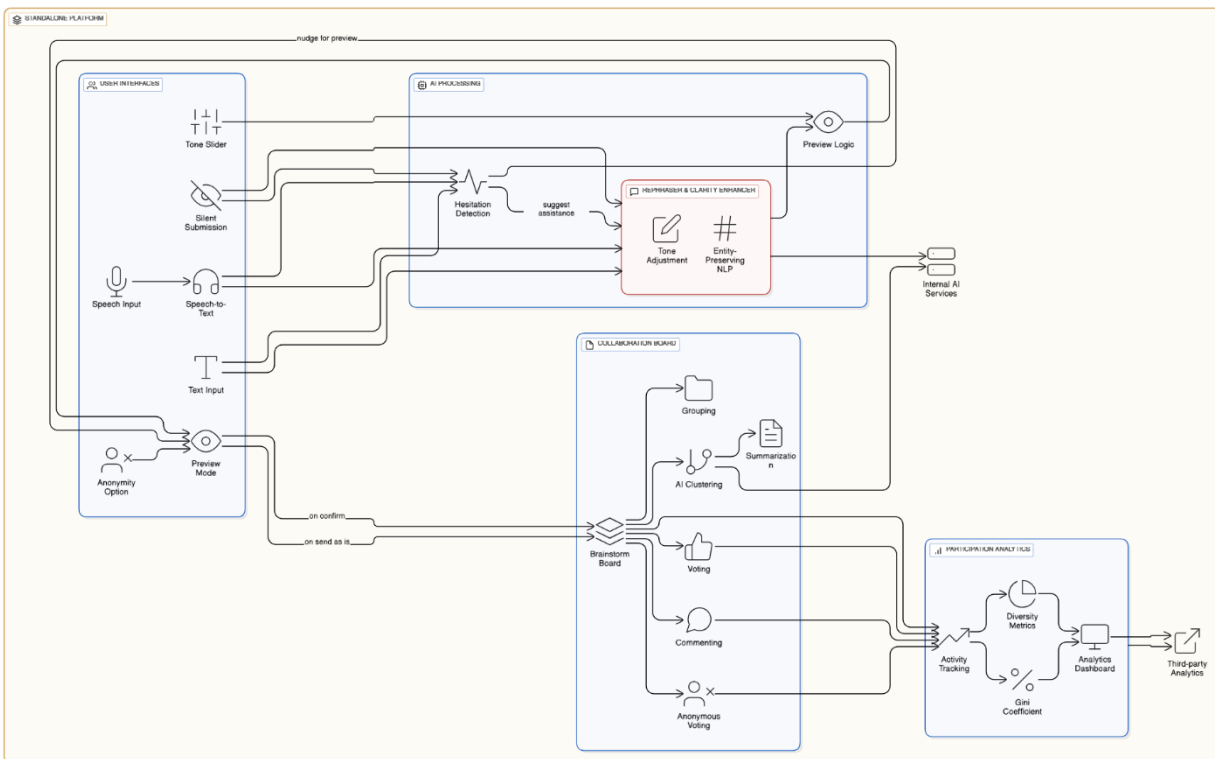
This objective seeks to build an analytics module that measures and visualizes participation diversity during brainstorming sessions. It will provide insights into the number of contributors, balance between frequent and silent participants, and overall fairness of idea-sharing. The analytics will be displayed through an interactive dashboard, giving Scrum Masters and team leaders the ability to monitor inclusivity and identify areas where participation can be improved.

Methodology

System Architecture



Component Architecture



4.0 Project Requirements

4.1 Functional Requirements

1. User Input & Submission
 - Support idea submission via text.
 - Allow anonymous or silent submission of ideas.
 - Enable optional speech-to-text for verbal contributors.
2. AI Rephraser & Confidence-Booster
 - Rephrase raw or fragmented inputs into clear, grammatically correct statements.
 - Preserve technical entities (e.g., IDs, numbers, technical terms).
 - Provide a preview function showing the original and rephrased version.
 - Allow tone adjustment (polite, neutral, assertive).
3. Hesitation Detection & Nudging
 - Detect incomplete or abandoned inputs (text or speech).
 - Prompt users with supportive nudges (e.g., “Would you like me to help rephrase this idea?”).
4. Idea Pool & Organization
 - Collect all approved ideas in a shared digital workspace.
 - Group similar contributions using AI clustering.
 - Provide automated summaries for clustered ideas.
5. Inclusivity Analytics
 - Track number of contributors and their participation balance.
 - Generate diversity metrics (e.g., share of inputs from silent vs. frequent contributors).
 - Display results in a dashboard with visual graphs.
 - Allow exporting inclusivity reports for review by team leads.

4.2 Non-Functional Requirements

- **Performance:**
Process documents in under 10 seconds; generate forecasts in real-time (<5 seconds).
- **Scalability:**
Handle up to 100 concurrent small-scale projects without degradation.
- **Usability:**
Intuitive mobile/web interface with support for Sinhala/English languages.
- **Reliability:**
99% uptime, with offline capabilities for remote sites.
- **Accuracy:**
>90% in material predictions and defect detections.
- **Interoperability:**
Integrate with existing BIM tools and supply chain APIs.

4.3 System Requirements

1. Hardware Requirements:

- **Development:** High-end PC with a GPU (e.g., NVIDIA RTX 3060 or equivalent, 16GB RAM) for training machine learning models and processing large datasets.
- **Deployment:** Cloud servers (e.g., AWS EC2 instances with at least 8 vCPUs, 32GB RAM, and GPU support) for scalable real-time processing and storage.
- **Support: Server-Side:**
 - High-availability server clusters with redundant GPUs (e.g., NVIDIA A100, 40GB RAM per node) and SSD storage (minimum 1TB) to handle peak loads and ensure uptime for real-time data processing.
 - Network infrastructure with a minimum bandwidth of 100 Mbps and load balancers for distributed computing across multiple nodes.

- **Support: Client-Side (Web & Mobile App):**

- Mobile devices with quad-core processors (e.g., Snapdragon 720G or A13 Bionic), 4GB RAM, and 64GB storage for smooth app performance on Android 10+ or iOS 14+.
- Web clients requiring modern browsers (e.g., Chrome 90+, Firefox 78+) on devices with at least 8GB RAM and 2GHz dual-core CPU for efficient dashboard rendering.
- Optional: Tablets (e.g., iPad Air with 4GB RAM) for enhanced on-site data entry and visualization.

2. Software Requirements:

- **Support: Server-Side:**

- **Operating System:** Linux (e.g., Ubuntu 20.04 LTS) for stability and compatibility with AI frameworks.
- **Programming Language:** Python 3.10+ with libraries like TensorFlow, PyTorch, and Hugging Face Transformers for NLP and ML.
- **Database:** PostgreSQL for structured data (e.g., user inputs, forecasts) and MongoDB for unstructured data (e.g., documents, logs).
- **Framework:** Flask or Django for backend API development, Docker for containerization, and Git for version control.
- **Cloud Services:** AWS or Google Cloud for hosting, with Firebase for real-time storage and synchronization.

- **Client-Side (Web & Mobile App):**

- Web: React.js with Node.js for a responsive browser interface, supporting Sinhala/English languages.
- Mobile App: React Native for cross-platform (Android/iOS) development, with offline capabilities and voice input support.
- APIs: RESTful APIs for secure data exchange between client and server, using HTTPS.
- Tools: Android Studio and Xcode for mobile deployment, with Chrome DevTools for web debugging.

3. Network Requirements:

- **Bandwidth:** Minimum 10 Mbps internet connection for cloud synchronization and real-time data updates, with support for low-bandwidth modes (e.g., 2G/3G) for remote sites.
- **Latency:** Maximum 200 ms round-trip latency for real-time alerts and app responsiveness, ensuring seamless user experience.
- **Reliability:** 99.9% uptime with failover mechanisms (e.g., redundant ISPs) to handle network disruptions in rural Sri Lankan construction sites.
- **Protocols:** Secure HTTPS for all data transmission, with WebSocket support for real-time updates between the Mobile App and server.
- **Coverage:** Support for mobile data networks (e.g., 4G LTE) and Wi-Fi, with offline caching capabilities for areas with intermittent connectivity.

4. Security Requirements:

5. Maintenance and Support:

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